

Lecture 3 Task — Open-Ended App or Game Development

Project Title: Personalized Trainer

App Idea: Users enter their current weight, desired weight, age, height, gender, and activity level, and select whether they want to gain or lose weight. Based on these inputs, the app generates a fully personalized plan including a 7-day meal schedule with calorie and macro targets, a weekly workout split, and expert tips — all tailored to the user's goal and preferred training environment (home or gym).

Purpose of the Project: The purpose of this project is to provide users with a personalized fitness and nutrition guide without the need for a professional trainer or dietitian. It simplifies the process of body transformation by giving users a clear, actionable plan based on their individual stats and goals.

Which AI Tools Were Used: Claude (claude.ai) — used for both generating the application code and powering the in-app AI plan generation.

Initial Prompt Used: *"Help me create an application for weight gain or weight loss. First give me 5 ideas on what type of application can be made. Don't explain deeply. Just one line each."*

Prompt Iterations Used During Development:

1. **Idea Generation:** Asked Claude for 5 one-line app ideas related to weight management to explore options before committing to one direction.
2. **Core App Development:** Selected the Meal and Workout Scheduler idea and prompted Claude to build a full application where users input their stats and goal, and receive a personalized diet and workout plan. Specified that the UI should be professional, user-friendly, and deployable on Netlify.
3. **Feature Addition & Bug Fix:** Requested two improvements — adding a workout environment selector (bodyweight at home vs. gym with equipment), and fixing a bug where citation tags were rendering visibly inside the generated plan content.

What Worked Well: The iterative prompting technique and think hard technique worked very effectively throughout the development process. Starting with broad idea generation and then narrowing down to specific requirements allowed the development to stay focused. Claude was

able to generate a fully functional, professionally designed application in just three prompt iterations.

What Did Not Work: The overall development process was smooth. No major functionality issues were encountered. The app was built end-to-end in approximately 5 minutes of active prompting.

Bugs or Challenges Faced: One bug was identified after the initial build — citation-style tags (such as ``) generated by the AI were rendered as visible text inside the workout and meal plan sections, cluttering the output and making it look unprofessional.

How the Issues Were Solved: The bug was resolved through a follow-up prompt describing the issue clearly. Claude fixed it by adding an HTML sanitization function that strips unwanted tags from AI-generated content before it renders on screen, and by tightening the internal system prompt to explicitly prevent such tags from being generated in the first place.

Features Added or Improved: In the initial version, the app assumed a default workout type when generating the plan. After reviewing the output, a third iteration was used to add a workout environment toggle — allowing users to explicitly choose between **Bodyweight training** (no equipment, home-friendly) and **Gym training** (barbells, machines, cables). This made the generated plans significantly more relevant and practical for each user.

Final Result: A fully deployed, AI-powered web application accessible at: <https://personalizedtrainer.netlify.app> 

Does the Project Solve a Real-World Problem? Yes. Access to personal trainers and nutritionists is expensive and not available to everyone. This app democratizes that guidance by generating personalized, science-backed fitness and diet plans instantly and for free — making it genuinely useful for students, beginners, and anyone looking to improve their health without professional support.